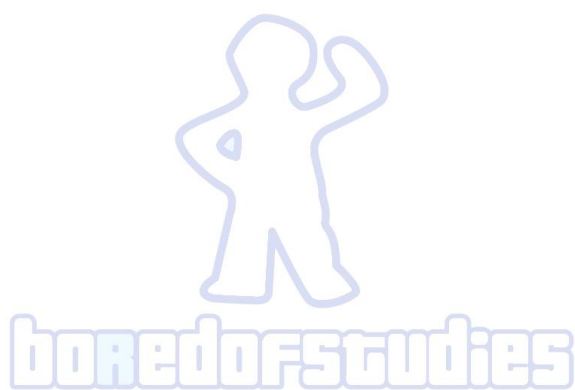




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2023

BORED OF STUDIES TRIAL EXAMINATION

3rd October

Mathematics Extension 1

**General
instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using a black or blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

**Total marks:
70**

Section I – 10 marks (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 5–10)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Consider the following table of values for a function $f(x)$.

x	$f(x)$	$f'(x)$
0	-1	1
1	0	2
2	2	3

Suppose that $f(x)$ has a unique inverse function $g(x)$ for all real x .
What is the value of $g'(0)$?

- (A) -1 (B) 1
(C) $\frac{1}{3}$ (D) $\frac{1}{2}$
- 2 When the polynomial $P(x) = acx^3 + bcx + d$ is divided by $D(x)$, it gives cx with a remainder of d . What is $D(x)$?
- (A) $-ax^2 - b$ (B) $-ax^2 + b$
(C) $ax^2 - b$ (D) $ax^2 + b$
- 3 What is the range of $y = \sin^{-1}(1 - \sqrt{1 + 2x^2})$
- (A) $\left[-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right]$ (B) $\left[-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}}\right]$ (C) $\left[-\frac{\pi}{2}, 0\right]$ (D) $\left[0, \frac{\pi}{2}\right]$
- 4 Suppose that the vectors $a\mathbf{i} + b\mathbf{j}$ and $c\mathbf{i} + d\mathbf{j}$ are perpendicular. Which of the following *could* be true?
- (A) $a < 0, b < 0, c < 0$ and $d < 0$ (B) $a < 0, b < 0, c > 0$ and $d > 0$
(C) $a > 0, b < 0, c > 0$ and $d < 0$ (D) $a > 0, b > 0, c < 0$ and $d > 0$

- 5 Let $(3x + 1)^8 = c_0 + c_1x + c_2x^2 + \cdots + c_8x^8$ for some constants $c_0, c_1, c_2, \dots, c_7$ and c_8 . Which of the following has the largest value?

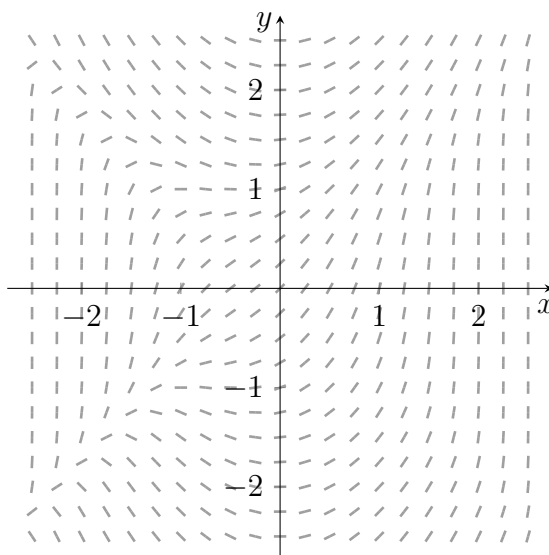
- (A) c_1 (B) c_3
(C) c_5 (D) c_7

- 6 Suppose that $\sqrt{2}\sin x + \cos\left(x + \frac{\pi}{4}\right) + R\cos(x + \alpha) = 0$ where $R > 0$ and $0 < \alpha < 2\pi$.

What is the value of α ?

- (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{7\pi}{4}$

- 7 Which of the following differential equations best represents the direction field below?



- (A) $\frac{dy}{dx} = x + e^{x^2+y^2}$ (B) $\frac{dy}{dx} = x + e^{-x^2+y^2}$
(C) $\frac{dy}{dx} = x + e^{x^2-y^2}$ (D) $\frac{dy}{dx} = x + e^{-x^2-y^2}$

- 8 A group of 6 people have reserved seating tickets for a concert. The six seats are split into 4 adjacent seats in one row and the remaining 2 adjacent seats in another row.

There is a couple in the group who must sit next to each other. How many possible seating arrangements are there for the group?

- (A) 48 (B) 120 (C) 144 (D) 192

- 9 Let $f(x) = 8 \cos^6 x - 12 \cos^4 x + 6 \cos^2 x - 1$. Which expression is equal to $\int f(x) dx$?

(A) $\frac{1}{24} \sin 6x + \frac{3}{8} \sin 2x + c$

(B) $\frac{1}{12} \sin 6x + \frac{3}{4} \sin 2x + c$

(C) $\frac{(\sin 2x)^4}{4} + c$

(D) $\frac{(\sin 2x)^4}{8} + c$

- 10 Suppose that a polynomial $P(x)$ is such that there exists some $a > 0$ where

$$\int_0^a P(x) dx = 0.$$

Which of the following polynomials satisfies this condition?

(A) $-x(x^2 + 1)$

(B) $-x(x - 1)(x - 4)$

(C) $(x + 1)(x - 1)(x - 2)$

(D) $(x + 1)(x + 2)(x + 3)$

Section II

60 marks

Attempt Questions 11—14

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet

- (a) Find the projection of the vector $\underline{i} - 2\underline{j}$ onto the line $2x + 3y + 6 = 0$. **2**
- (b) Show that $\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8}$. **1**
- (c) The polynomial $P(x) = x^3 - 3x + 2$ has a double root.
- (i) Find all the roots of $P(x)$. **2**
- (ii) Hence, or otherwise, solve $\frac{x^2 - x + 2}{4x} \leq \frac{1}{x + 1}$. **3**
- (d) A circular metallic disc initially has a radius of 1 cm. As it gets heated, the rate at which its radius R expands at time t is given by
- $$\frac{dR}{dt} = 10 - 2R.$$
- (i) Find the specific solution to the differential equation of R in terms of t . **1**
- (ii) When is the growth rate of disc's **area** the fastest? **2**
- (e) A boat is sailing at constant velocity in a straight line across a lake. At time t hours, its displacement in kilometres is given by the position vector $\underline{r}$. Let the unit vectors $\underline{i}$ and $\underline{j}$ be directed at due east and due north respectively. The boat is initially at $-5\underline{i} + 10\underline{j}$ and 3 hours later it is at $4\underline{i} - 2\underline{j}$.
- (i) Find the displacement equation $\underline{r}$ of the boat in terms of t . **2**
- (ii) Find the velocity of the boat in terms of magnitude and direction in true bearings (to the nearest degree). **2**

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet

- (a) A multiple choice test of 25 questions is designed such that there are four possible options and only one correct answer for each question. Each correct answer gains 3 marks but each incorrect answer loses 1 mark.

Suppose a certain student chooses to make a non-serious attempt and answers all 25 questions at random.

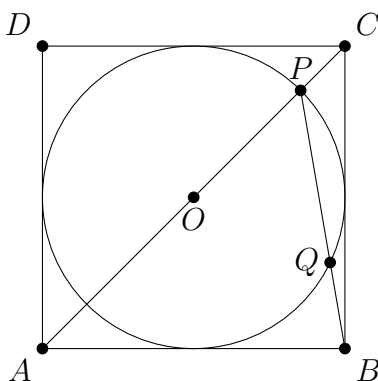
- (i) Find the probability the student will get exactly 15 marks in total. **1**

- (ii) Now suppose that for each question, the student can choose to make a serious attempt (not answering at random) or a non-serious attempt (answering at random), both of which are equally likely. **2**

If the student makes a serious attempt at the question, the probability that they get the question correct is $\frac{5}{12}$.

By considering a normal approximation, find the probability that the student will score at least 27 marks.

- (b) A circle with centre O is inscribed in a square $ABCD$ with a side length of 1 unit as shown in the diagram below. Let P be the point on the circle where the diagonal AC intersects the circle and is closest to C . Let Q be the point where PB intersects the circle. Suppose that the vectors representing $\overrightarrow{AB}$ and $\overrightarrow{AD}$ are $\hat{i}$ and $\hat{j}$ respectively.



- (i) Show that the vector representing $\overrightarrow{BP}$ is $\left(\frac{\sqrt{2}-2}{4}\right)\hat{i} + \left(\frac{\sqrt{2}+2}{4}\right)\hat{j}$. **2**

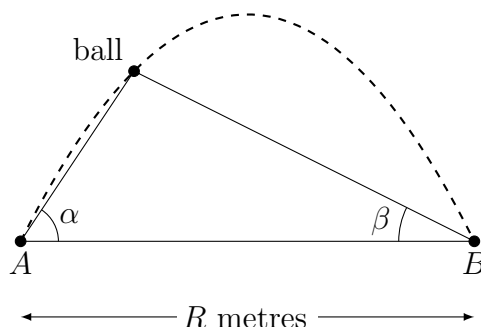
- (ii) Hence, show that $\cos \angle POQ = \frac{1}{3}$. **2**

Question 12 continues on page 7

Question 12 (continued)

- (c) Student A launches a ball into the air from the ground at an angle of θ to the horizontal and an initial speed of u . Student B is on the ground R metres away from student A and will catch the ball.

Whilst the ball is in flight, student A and student B are both stationary and view the ball at an angle of elevation of α and β respectively at time t .



The displacement vector of the ball relative to the origin at student A is given by

$$\underline{r} = ut \cos \theta \underline{i} + \left(ut \sin \theta - \frac{gt^2}{2} \right) \underline{j} \quad \text{(Do NOT prove this)}$$

where g is the acceleration due to gravity.

- (i) Show that $\tan \alpha$ decreases linearly over time. 1
- (ii) Show that $\tan \alpha + \tan \beta$ is independent of time. 2
- (d) It can be shown that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$. (Do NOT prove this)
- (i) Use this result to show that 2
- $$\sin 7\theta + \sin \theta = 8 \sin \theta - 56 \sin^3 \theta + 112 \sin^5 \theta - 64 \sin^7 \theta.$$
- (ii) Hence, find the roots of $64x^6 - 112x^4 + 56x^2 - 7$. 1
- (iii) Deduce that 2

$$\operatorname{cosec}^2 \frac{\pi}{7} + \operatorname{cosec}^2 \frac{2\pi}{7} + \operatorname{cosec}^2 \frac{3\pi}{7} = 8.$$

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet

- (a) By considering an appropriate t-formula, or otherwise, show that **2**

$$\tan^{-1} x = \begin{cases} -\frac{1}{2} \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) & \text{for } x < 0 \\ \frac{1}{2} \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) & \text{for } x \geq 0. \end{cases}$$

- (b) Let (x, y) be a point on the circle $x^2 + y^2 = r^2$ for some constant $r > 0$.
Let $f(z)$ be a probability density function on all real z , with a global maximum turning point at $(0, 1)$.

Suppose that $f(z)$ has the property that for some values x and y on the circle

$$g(r) = f(x)f(y)$$

where $g(r)$ is some function of r , independent of x and y .

- (i) By considering the parametric equations of the circle, show that for $x \neq 0$ and $y \neq 0$ **2**

$$\frac{f'(x)}{xf(x)} = \frac{f'(y)}{yf(y)}.$$

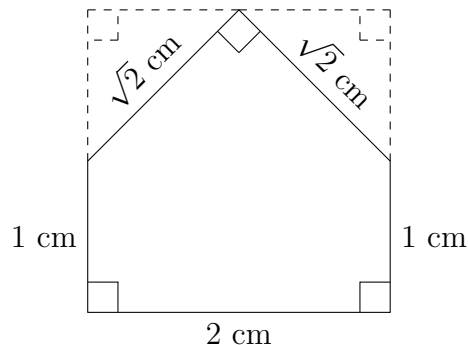
- (ii) Explain why $\frac{f'(x)}{xf(x)} = -k$ for some constant $k > 0$. **2**

- (iii) Hence, use the substitution $u = x\sqrt{k}$ to find a specific solution to the differential equation in part (ii). **3**

Question 13 continues on page 9

Question 13 (continued)

- (c) Consider a pentagon with side lengths as shown in the diagram below. Note that the top vertex bisects the side length of a 2 cm by 2 cm square.



- (i) Explain why the longest distance between any two points within the pentagon must be between two non-adjacent vertices. 1
- (ii) Hence, show that the longest distance between any two points within the pentagon is $\sqrt{5}$ cm. 1
- (iii) Six points are randomly placed within a 3 cm by 4 cm rectangle. Use the pigeonhole principle and the result of part (ii) to show that there exists at least one pair of points that are within $\sqrt{5}$ cm of each other. 1
- (d) Prove by mathematical induction that for positive integers n 3

$$\frac{\sin x}{\cos x + \cos 3x} + \frac{\sin x}{\cos x + \cos 5x} + \cdots + \frac{\sin x}{\cos x + \cos(2n+1)x} = \frac{\tan(n+1)x - \tan x}{2}.$$

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet

- (a) (i) By considering the graph of $y = x^2 - 3x + 3$, or otherwise, sketch the graph of $y = \frac{1}{x^2 - 3x + 3}$, showing any turning points and asymptotes. **2**

- (ii) On the same set of axes, sketch the graph of $y = \tan^{-1} \left(\frac{1}{x^2 - 3x + 3} \right)$. **1**

- (iii) Explain why the graphs of $y = \frac{1}{x^2 - 3x + 3}$ and $y = \tan^{-1} \left(\frac{1}{x^2 - 3x + 3} \right)$ do not have a point of intersection. **1**

- (iv) Differentiate $(x + c) \tan^{-1}(x + c) - \ln \left(\sqrt{1 + (x + c)^2} \right)$ with respect to x for some constant c . **1**

- (v) Hence, use the expansion of $\tan(A + B)$ to find the area between the curves **4**

$$y = \frac{1}{x^2 - 3x + 3} \quad \text{and} \quad y = \tan^{-1} \left(\frac{1}{x^2 - 3x + 3} \right)$$

for $1 \leq x \leq 2$.

- (b) A soccer player performs a training drill where she needs to score exactly r goals. Her training session is complete after she scores her r^{th} goal. **2**

She is given a maximum of n attempts and manages to score r goals and complete her training session. By considering two different methods to count the number of ways she can do this, show that

$$\binom{r-1}{r-1} + \binom{r}{r-1} + \binom{r+1}{r-1} + \cdots + \binom{n-1}{r-1} = \binom{n}{r}.$$

- (c) (i) Show that for integer $n \geq 4$ **1**

$$n^4 = 24 \binom{n}{4} + 36 \binom{n}{3} + 14 \binom{n}{2} + \binom{n}{1}.$$

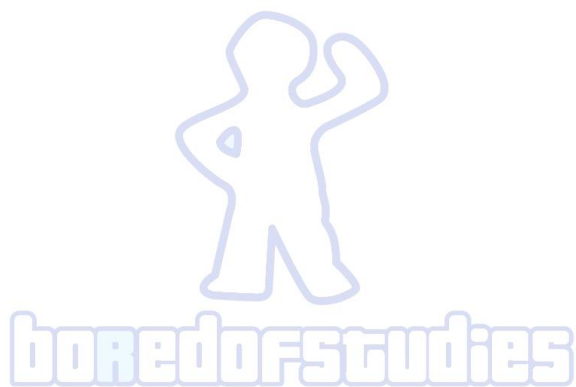
- (ii) Hence, using the results in parts (b) and (c)(i), show *without* induction that **3**

$$1^4 + 2^4 + 3^4 + \cdots + n^4 = \frac{1}{30} n(n+1)(2n+1)(3n^2 + 3n - 1).$$

End of paper

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2023

BORED OF STUDIES TRIAL EXAMINATION

Mathematics Extension 1

Solutions

Section I

Answers

1 D

6 B

2 D

7 C

3 C

8 D

4 D

9 A

5 D

10 B

Brief explanations

1 If $y = f^{-1}(x)$ then

$$\begin{aligned}x &= f(y) \\ \frac{dx}{dy} &= f'(y) \\ \frac{dy}{dx} &= \frac{1}{f'(f^{-1}(x))} \quad \text{let } g(x) = f^{-1}(x) \\ g'(x) &= \frac{1}{f'(g(x))} \\ g'(0) &= \frac{1}{f'(g(0))} \quad \text{but } g(0) = 1 \\ &= \frac{1}{f'(1)} \\ &= \frac{1}{2}\end{aligned}$$

Hence, the answer is (D).

2 Given the divisor and remainder

$$\begin{aligned}P(x) &= D(x)cx + d \\ acx^3 + bcx + d &\equiv D(x)cx + d \\ cx(ax^2 + b) &\equiv D(x)cx + d \\ D(x) &= ax^2 + b\end{aligned}$$

Hence, the answer is (D).

- 3 Since $x^2 \geq 0$ and $\sin^{-1} x$ is an increasing function then

$$\begin{aligned} 1 + 2x^2 &\geq 1 \\ -\sqrt{1 + 2x^2} &\leq -1 \\ 1 - \sqrt{1 + 2x^2} &\leq 0 \\ \sin^{-1} \left(1 - \sqrt{1 + 2x^2} \right) &\leq 0 \end{aligned}$$

However, note that the lower bound of $1 - \sqrt{1 + 2x^2}$ must be -1 so the range must be $\left[-\frac{\pi}{2}, 0\right]$. Hence, the answer is (C).

- 4 The dot product must be zero for the vectors to be perpendicular so $ac + bd = 0$. A necessary condition is that ac and bd must be opposite in sign. The only option which has this property is (D).

- 5 The general term is $c_k = \binom{8}{k} 3^k$ so numerically evaluating each option

$$\begin{aligned} c_1 &= \binom{8}{1} \times 3^1 = 24 \\ c_3 &= \binom{8}{3} \times 3^3 = 1512 \\ c_5 &= \binom{8}{5} \times 3^5 = 13608 \\ c_7 &= \binom{8}{7} \times 3^7 = 17496 \end{aligned}$$

Hence, the answer is (D).

Remark: Note that this is actually *not* the largest coefficient of the entire binomial expansion. However, it is the largest coefficient of the given options.

- 6 Rearranging gives

$$\begin{aligned} R \cos(x + \alpha) &= -\sqrt{2} \sin x - \cos \left(x + \frac{\pi}{4} \right) \\ R \cos x \cos \alpha - R \sin x \sin \alpha &= -\sqrt{2} \sin x - \frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x \\ &= -\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x \\ R \cos \alpha &= -\frac{1}{\sqrt{2}} \quad \text{and} \quad R \sin \alpha = \frac{1}{\sqrt{2}} \\ \tan \alpha &= -1 \\ \alpha &= \frac{3\pi}{4} \end{aligned}$$

Noting that since $R > 0$ then the only quadrant which has positive value for $\sin \alpha$ and negative value for $\cos \alpha$ is the second quadrant. Hence, the answer is (B).

- 7 First consider the case when $x = 0$ then as y increases to a large positive value, the slope is almost flat. This similarly occurs when y decreases to a large negative value. The only options which have this property are (C) and (D).

Notice there is a flat slope at $(-1, 1)$. Option (C) gives $\frac{dy}{dx} = 0$ at this point whereas option (D) gives $\frac{dy}{dx} = -1 + e^{-2}$ which is negative at this point.

Only option (C) is consistent with this observation.

- 8 There are two cases to consider.

Case 1: The couple sit in the two seat row

There are $2!$ ways for the couple to arrange in the two seat row. There are $4!$ ways to arrange in the four seat row. The total number of arrangements is $4! \times 2!$.

Case 2: The couple sit in the four seat row

In the two seat row there are $\binom{4}{2}$ possible groups and $2!$ ways to arrange within that row. In the four seat row, there are $2!$ ways for the couple to arrange within themselves. Treating the couple as one group there are $3!$ ways to arrange all members of the four seat row. Therefore, the total number of arrangements is $\binom{4}{2} \times 2! \times 2! \times 3!$.

In total there are $4! \times 2! + \binom{4}{2} \times 2! \times 2! \times 3!$, or equivalently, 192 ways to arrange the group. Hence, the answer is (D).

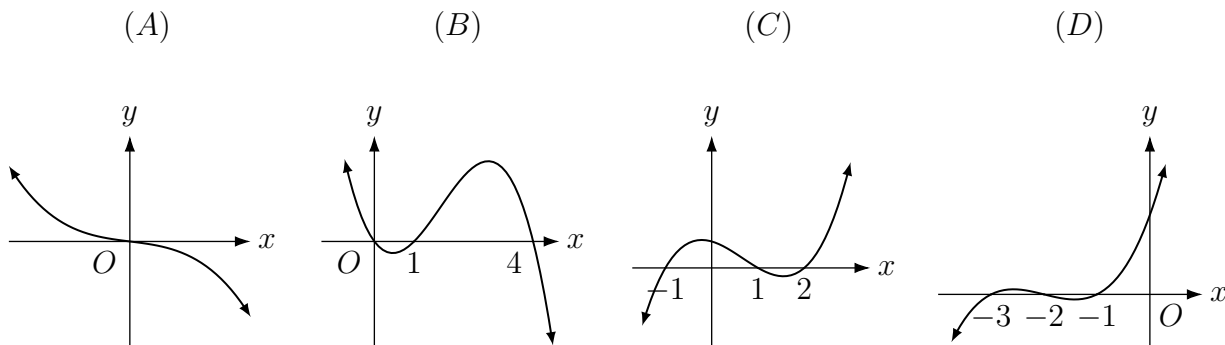
- 9 Notice that

$$\begin{aligned}
 f(x) &= 8 \cos^6 x - 12 \cos^4 x + 6 \cos^2 x - 1 \\
 &= 8 \cos^6 x - 8 \cos^4 x + 2 \cos^2 x - 4 \cos^4 x + 4 \cos^2 x - 1 \\
 &= 2 \cos^2 x (4 \cos^4 x - 4 \cos^2 x + 1) - (4 \cos^4 x - 4 \cos^2 x + 1) \\
 &= (2 \cos^2 x - 1)(4 \cos^4 x - 4 \cos^2 x + 1) \\
 &= (2 \cos^2 x - 1)^3 \\
 &= \cos 2x \cos^2 2x \\
 &= \frac{1}{2} \cos 2x (\cos 4x + 1) \\
 &= \frac{1}{4} (\cos(4x - 2x) + \cos(4x + 2x)) + \frac{1}{2} \cos 2x \\
 &= \frac{1}{4} \cos 6x + \frac{3}{4} \cos 2x \\
 \int f(x) dx &= \int \left(\frac{1}{4} \cos 6x + \frac{3}{4} \cos 2x \right) dx \\
 &= \frac{1}{24} \sin 6x + \frac{3}{8} \sin 2x + c
 \end{aligned}$$

Hence, the answer is (A).

- 10 A necessary condition for the integral to be zero is that the polynomial must have a mix of positive and negative function values in the domain $x > 0$.

Consider the sketches of each option below.



$$y = -x(x^2 + 1) \quad y = -x(x-1)(x-4) \quad y = (x+1)(x-1)(x-2) \quad y = (x+1)(x+2)(x+3)$$

For option (A), when $x > 0$ then $-x(x^2 + 1) < 0$ so it is not possible for the integral to be zero.

For option (D), when $x > 0$ then $(x+1)(x+2)(x+3) > 0$ so it is not possible for the integral to be zero.

For option (C), it can be seen that $\int_0^1 (x+1)(x-1)(x-2) dx > 0$. So now consider

$$\begin{aligned} \int_0^2 (x+1)(x-1)(x-2) dx &= \int_0^2 (x^3 - 2x^2 - x + 2) dx \\ &= \left[\frac{x^4}{4} - \frac{2x^3}{3} - \frac{x^2}{2} + 2x \right]_0^2 \\ &= \frac{2}{3} \end{aligned}$$

This means that the entire ‘negative’ region is insufficient to offset the ‘positive’ region, so the answer cannot be (C).

For option (B), it can be seen that $\int_0^1 -x(x-1)(x-4) dx < 0$. So now consider

$$\begin{aligned} \int_0^4 -x(x-1)(x-4) dx &= \int_0^4 (-x^3 + 5x^2 - 4x) dx \\ &= \left[-\frac{x^4}{4} + \frac{5x^3}{3} - 2x^2 \right]_0^4 \\ &= \frac{32}{3} \end{aligned}$$

This means that the ‘positive’ region is more than sufficient enough to offset the ‘negative’ region. This means there must exist some $1 < a < 4$ where $\int_0^a -x(x-1)(x-4) dx = 0$. Hence, the answer is (B).

Question 11

- (a) First find a vector parallel to the line $2x + 3y + 6 = 0$. Use the x -intercept $(-3, 0)$ and y -intercept $(0, -2)$ as convenient points which can construct a vector $\underline{b} = 3\underline{i} - 2\underline{j}$. Let $\underline{a} = \underline{i} - 2\underline{j}$ and so

$$\begin{aligned}\text{proj}_{\underline{b}}\underline{a} &= (\underline{a} \cdot \hat{\underline{b}})\hat{\underline{b}} \\ &= \frac{1}{\sqrt{13}}(\underline{i} - 2\underline{j}) \cdot (3\underline{i} - 2\underline{j})\hat{\underline{b}} \\ &= \frac{7}{\sqrt{13}}\hat{\underline{b}} \\ &= \frac{7}{13}(3\underline{i} - 2\underline{j})\end{aligned}$$

- (b) Using the fact that $\sin 2x = 2 \sin x \cos x$

$$\begin{aligned}\text{LHS} &= \cos 20^\circ \cos 40^\circ \cos 80^\circ \\ &= \frac{\sin 40^\circ}{2 \sin 20^\circ} \times \frac{\sin 80^\circ}{2 \sin 40^\circ} \times \frac{\sin 160^\circ}{2 \sin 80^\circ} \\ &= \frac{\sin 160^\circ}{8 \sin 20^\circ} \quad \text{but } \sin(180^\circ - x) = \sin x \\ &= \frac{1}{8} \\ &= \text{RHS}\end{aligned}$$

- (c) (i) Since $P(x)$ has a double root then this double root satisfies $P'(x) = 0$

$$\begin{aligned}3x^2 - 3 &= 0 \\ x &= \pm 1\end{aligned}$$

$P(-1) = 4$ and $P(1) = 0$ so the double root is $x = 1$. Let the remaining root be α . By sum of roots

$$\begin{aligned}1 + 1 + \alpha &= 0 \\ \alpha &= -2\end{aligned}$$

The roots of $P(x)$ are $1, 1, -2$.

(ii) Solving the inequality

$$\begin{aligned}
 \frac{x^2 - x + 2}{4x} &\leq \frac{1}{x + 1} \\
 \frac{x^2 - x + 2}{4x} - \frac{1}{x + 1} &\leq 0 \\
 \frac{(x + 1)(x^2 - x + 2) - 4x}{4x(x + 1)} &\leq 0 \\
 \frac{x^3 - x^2 + 2x + x^2 - x + 2 - 4x}{x(x + 1)} &\leq 0 \\
 x(x + 1)(x^3 - 3x + 2) &\leq 0 \\
 x(x + 1)(x - 1)^2(x + 2) &\leq 0 \quad \text{by factor theorem using roots from part (i)}
 \end{aligned}$$

Since $(x - 1)^2 \geq 0$ for all real x then $x = 1$ satisfies the inequality. When $x \neq 1$ then $(x - 1)^2 > 0$ which implies $x(x + 1)(x + 2) \leq 0$.

Noting that $x \neq 0$ and $x \neq -1$, the solutions are $x = 1, x \leq -2$ or $-1 < x < 0$.

(d) (i) Note that $\frac{dR}{dt} = -2(R - 5)$.

The general solution to the shifted exponential decay equation is $R = 5 + Be^{-2t}$ for some constant B .

When $t = 0, R = 1$ so $B = -4$.

Hence, the specific solution is $R = 5 - 4e^{-2t}$.

(ii) Let A be the area of the disc so $A = \pi R^2$ so using related rates

$$\begin{aligned}
 \frac{dA}{dt} &= \frac{dA}{dR} \times \frac{dR}{dt} \\
 &= 4\pi R(5 - R)
 \end{aligned}$$

The rate of change in the disc's area is described by the concave down parabola $y = 4\pi x(5 - x)$. This has a global maximum when $x = \frac{5}{2}$. Substitute this into R to find t

$$\begin{aligned}
 \frac{5}{2} &= 5 - 4e^{-2t} \\
 e^{-2t} &= \frac{5}{8} \\
 t &= \frac{1}{2} \ln \frac{8}{5}
 \end{aligned}$$

- (e) (i) Let the velocity vector be $\underline{v} = v_x \underline{i} + v_y \underline{j}$. Since the velocity is constant then at time t then

$$\begin{aligned}\underline{r} &= -5\underline{i} + 10\underline{j} + t(v_x \underline{i} + v_y \underline{j}) \\ &= (-5 + tv_x)\underline{i} + (10 + tv_y)\underline{j}\end{aligned}$$

When $t = 3$, $\underline{r} = 4\underline{i} - 2\underline{j}$ so

$$\begin{aligned}-5 + (3)v_x &= 4 \Rightarrow v_x = 3 \\ 10 + (3)v_y &= -2 \Rightarrow v_y = -4\end{aligned}$$

Hence, the displacement vector is $\underline{r} = (-5 + 3t)\underline{i} + (10 - 4t)\underline{j}$.

- (ii) From the working in part (i), the velocity vector is $\underline{v} = 3\underline{i} - 4\underline{j}$. It's magnitude is $|\underline{v}| = \sqrt{3^2 + 4^2} = 5$.

The direction of the vector in true bearing is the angle the vector makes to the positive vertical (as north).

The vector $3\underline{i} - 4\underline{j}$ lies in the fourth quadrant so if θ is the angle to the positive horizontal then $\tan \theta = \frac{4}{3}$. The true bearing is $90^\circ + \tan^{-1} \frac{4}{3} \approx 143^\circ \text{T}$.

Hence, the velocity of the boat is 5km per hour directed at 143°T .

Question 12

- (a) (i) Let X be the total number questions the student answers correctly. Let M be the total marks attained by the student, which is related by

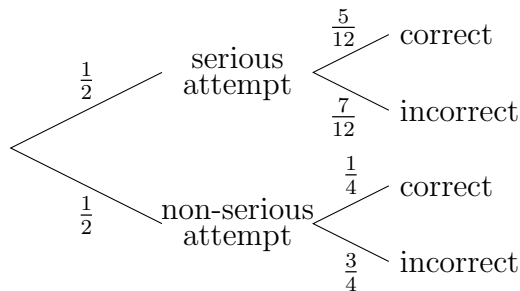
$$\begin{aligned} M &= 3X - (25 - X) \\ &= 4X - 25 \end{aligned}$$

Given $X \sim B(25, \frac{1}{4})$ then

$$\begin{aligned} P(M = 15) &= P(4X - 25 = 15) \\ &= P(X = 10) \\ &= \binom{25}{10} \left(\frac{1}{4}\right)^{10} \left(\frac{3}{4}\right)^{15} \end{aligned}$$

- (ii) Let S be a Bernoulli random variable which represents whether the student makes a serious or non-serious attempt at the question, so $S \sim B(1, \frac{1}{2})$.

Consider a probability tree of all the outcomes.



The unconditional probability of getting the answer correct for a question is

$$\begin{aligned} P(X = 1) &= P(X = 1|S = 0) + P(X = 1|S = 1) \\ &= \frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \times \frac{5}{12} \\ &= \frac{1}{3} \end{aligned}$$

The overall probability of getting the correct answer has improved to $\frac{1}{3}$ so now $X \sim B(25, \frac{1}{3})$ which means

$$\begin{aligned} P(M \geq 27) &= P(4X - 25 \geq 27) \\ &= P(X \geq 13) \\ &= P\left(\frac{X - 25(\frac{1}{3})}{\sqrt{25(\frac{1}{3})(\frac{2}{3})}} \geq \frac{13 - 25(\frac{1}{3})}{\sqrt{25(\frac{1}{3})(\frac{2}{3})}}\right) \\ &\approx P(Z \geq 1.98) \quad \text{where } Z \sim N(0, 1) \\ &\approx P(Z \geq 2) \\ &\approx 0.025 \end{aligned}$$

Noting that from the reference sheet, $P(-2 \leq Z \leq 2) = 0.95$, which implies $P(Z \geq 2) = 0.025$, by symmetry of the standard normal curve.

- (b) (i) Consider the vector $\overrightarrow{AP}$ which related to $\overrightarrow{AO}$ by a scalar. Let $\hat{u}$ be the unit vector in the direction of $\overrightarrow{AO}$.

$$\begin{aligned}\overrightarrow{AO} &= \frac{1}{2}\hat{i} + \frac{1}{2}\hat{j} \\ &= \frac{1}{2}(\hat{i} + \hat{j}) \\ \hat{u} &= \frac{1}{\sqrt{2}}(\hat{i} + \hat{j})\end{aligned}$$

Use this to derive $\overrightarrow{AP}$.

$$\begin{aligned}\overrightarrow{AP} &= |\overrightarrow{AP}|\hat{u} \\ &= \left(|\overrightarrow{AO}| + |\overrightarrow{OP}|\right)\hat{u} \quad \text{but } |\overrightarrow{OP}| = \frac{1}{2} \text{ and } |\overrightarrow{AO}| = \frac{1}{\sqrt{2}} \\ &= \frac{1}{\sqrt{2}}\left(\frac{1}{2} + \frac{1}{\sqrt{2}}\right)(\hat{i} + \hat{j}) \\ &= \frac{\sqrt{2} + 2}{4}(\hat{i} + \hat{j})\end{aligned}$$

Hence

$$\begin{aligned}\overrightarrow{BP} &= \overrightarrow{AP} - \overrightarrow{AB} \\ &= \frac{\sqrt{2} + 2}{4}(\hat{i} + \hat{j}) - \hat{i} \\ &= \left(\frac{\sqrt{2} - 2}{4}\right)\hat{i} + \left(\frac{\sqrt{2} + 2}{4}\right)\hat{j}\end{aligned}$$

(ii) First note that

$$\begin{aligned} OP &= OQ \quad (\text{equal radii}) \\ \therefore \triangle POQ &\text{ is isosceles} \\ \Rightarrow \angle OPQ &= \angle OQP \end{aligned}$$

Finding $\angle OPQ$ using the unit vector derived earlier in working for (i)

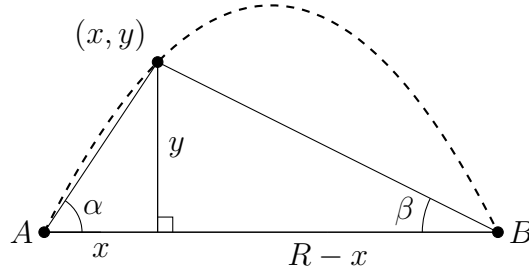
$$\begin{aligned} \cos \angle OPQ &= \frac{(-\hat{u}) \cdot \overrightarrow{PB}}{|\hat{u}| |\overrightarrow{PB}|} \\ &= \frac{\hat{u} \cdot \overrightarrow{BP}}{|\overrightarrow{BP}|} \\ \hat{u} \cdot \overrightarrow{BP} &= \frac{1}{\sqrt{2}} \left(\frac{\sqrt{2}-2}{4} \right) + \frac{1}{\sqrt{2}} \left(\frac{\sqrt{2}+2}{4} \right) \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} |\overrightarrow{BP}| &= \sqrt{\left(\frac{\sqrt{2}-2}{4} \right)^2 + \left(\frac{\sqrt{2}+2}{4} \right)^2} \\ &= \frac{1}{4} \sqrt{2 - 2\sqrt{2} + 4 + 2 + 2\sqrt{2} + 4} \\ &= \frac{\sqrt{3}}{2} \end{aligned}$$

Hence

$$\begin{aligned} \cos \angle OPQ &= \frac{1}{\sqrt{3}} \\ \cos \angle POQ &= \cos(\pi - 2\angle OPQ) \\ &= -\cos 2\angle OPQ \\ &= -(2\cos^2 \angle OPQ - 1) \quad \text{but } \cos \angle OPQ = \frac{1}{\sqrt{3}} \\ &= \frac{1}{3} \end{aligned}$$

- (c) (i) The displacement of the ball has coordinates (x, y) at time t as shown below.



$$\begin{aligned}\tan \alpha &= \frac{y}{x} \\ &= \frac{ut \sin \theta - \frac{gt^2}{2}}{ut \cos \theta} \\ &= \tan \theta - \frac{gt}{2u \cos \theta}\end{aligned}$$

Since $u > 0$, $g > 0$ and $0 < \theta < \frac{\pi}{2}$ then $\tan \alpha$ is a decreasing linear function of t .

- (ii) First note that the range R is a fixed value which occurs with some time of flight (call this T). This means that $R = uT \cos \theta$ and so

$$\begin{aligned}\tan \beta &= \frac{y}{R - x} \\ &= \frac{ut \sin \theta - \frac{gt^2}{2}}{uT \cos \theta - ut \cos \theta} \\ &= \frac{2ut \sin \theta - gt^2}{2u(T - t) \cos \theta}\end{aligned}$$

However, T can be evaluated directly by solving for $y = 0$

$$\begin{aligned}uT \sin \theta - \frac{gT^2}{2} &= 0 \\ T &= \frac{2u \sin \theta}{g} \quad \text{noting that } T > 0\end{aligned}$$

Noting that $t < T$ when the ball is in flight then

$$\begin{aligned}\tan \beta &= \frac{gtT - gt^2}{2u(T - t) \cos \theta} \\ &= \frac{gt(T - t)}{2u(T - t) \cos \theta} \\ &= \frac{gt}{2u \cos \theta} \\ \tan \alpha + \tan \beta &= \tan \theta - \frac{gt}{2u \cos \theta} + \frac{gt}{2u \cos \theta} \\ &= \tan \theta\end{aligned}$$

Hence, $\tan \alpha + \tan \beta$ is independent of time.

- (d) (i) Using sum to product and double angle results

$$\begin{aligned}
\text{LHS} &= \sin 7\theta + \sin \theta \\
&= \sin(4\theta + 3\theta) + \sin(4\theta - 3\theta) \\
&= 2 \sin 4\theta \cos 3\theta \\
&= 2(2 \sin 2\theta \cos 2\theta)(4 \cos^3 \theta - 3 \cos \theta) \quad \text{using the given result} \\
&= 8 \sin \theta \cos \theta (1 - 2 \sin^2 \theta)(4 \cos^3 \theta - 3 \cos \theta) \\
&= 8 \sin \theta \cos^2 \theta (1 - 2 \sin^2 \theta)(4 \cos^2 \theta - 3) \\
&= 8 \sin \theta (1 - \sin^2 \theta)(1 - 2 \sin^2 \theta)(4(1 - \sin^2 \theta) - 3) \\
&= 8 \sin \theta (1 - \sin^2 \theta)(1 - 2 \sin^2 \theta)(1 - 4 \sin^2 \theta) \\
&= 8 \sin \theta (1 - \sin^2 \theta)(1 - 6 \sin^2 \theta + 8 \sin^4 \theta) \\
&= 8 \sin \theta (1 - 6 \sin^2 \theta + 8 \sin^4 \theta - \sin^2 \theta + 6 \sin^4 \theta - 8 \sin^6 \theta) \\
&= 8 \sin \theta (1 - 7 \sin^2 \theta + 14 \sin^4 \theta - 8 \sin^6 \theta) \\
&= 8 \sin \theta - 56 \sin^3 \theta + 112 \sin^5 \theta - 64 \sin^7 \theta \\
&= \text{RHS}
\end{aligned}$$

- (ii) Rearranging the result in part (i) gives

$$\sin 7\theta = 7 \sin \theta - 56 \sin^3 \theta + 112 \sin^5 \theta - 64 \sin^7 \theta$$

Let $x = \sin \theta$, so the roots of $64x^6 - 112x^4 + 56x^2 - 7$ can be interpreted as solving the equation

$$\begin{aligned}
64 \sin^6 \theta - 112 \sin^4 \theta + 56 \sin^2 \theta - 7 &= 0 \\
-\frac{\sin 7\theta}{\sin \theta} &= 0 \\
\sin 7\theta &= 0 \quad \text{where } \sin \theta \neq 0 \\
7\theta &= \pm\pi, \pm 2\pi, \pm 3\pi \\
\theta &= \pm \frac{\pi}{7}, \pm \frac{2\pi}{7}, \pm \frac{3\pi}{7},
\end{aligned}$$

Note that other solutions (such as $\theta = 0$) either contradict the condition that $\sin \theta \neq 0$ or give the same value of $\sin \theta$ as the existing solutions.

The roots are $\pm \sin \frac{\pi}{7}$, $\pm \sin \frac{2\pi}{7}$ and $\pm \sin \frac{3\pi}{7}$.

- (iii) The 6th degree polynomial $64x^6 - 112x^4 + 56x^2 - 7$ can also be interpreted as a cubic polynomial by letting $u = x^2$, which reduces it to $64u^3 - 112u^2 + 56u - 7$.

Equivalently, this is a cubic polynomial in $u = \sin^2 \theta$.

From part (ii) it has the roots $\sin^2 \frac{\pi}{7}$, $\sin^2 \frac{2\pi}{7}$ and $\sin^2 \frac{3\pi}{7}$.

Let $\alpha = \sin^2 \frac{\pi}{7}$, $\beta = \sin^2 \frac{2\pi}{7}$ and $\gamma = \sin^2 \frac{3\pi}{7}$.

$$\begin{aligned}
 \text{LHS} &= \operatorname{cosec}^2 \frac{\pi}{7} + \operatorname{cosec}^2 \frac{2\pi}{7} + \operatorname{cosec}^2 \frac{3\pi}{7} \\
 &= \frac{1}{\sin^2 \frac{\pi}{7}} + \frac{1}{\sin^2 \frac{2\pi}{7}} + \frac{1}{\sin^2 \frac{3\pi}{7}} \\
 &= \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} \\
 &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\
 &= \frac{56}{64} \times \frac{64}{7} \\
 &= 8 \\
 &= \text{RHS}
 \end{aligned}$$

Question 13

- (a) First note that $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is an even function and when $x \rightarrow \infty$ then $\frac{1-x^2}{1+x^2} \rightarrow -1$ and when $x = 0$, $\frac{1-x^2}{1+x^2} = 1$.

This suggests that the domain of $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is $(-1, 1]$ and its range is $[0, \pi)$.

Let $x = \tan \theta \Rightarrow \theta = \tan^{-1} x$ with $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.

As a general solution, for some integer k

$$\begin{aligned}\cos 2\theta &= \frac{1-x^2}{1+x^2} \quad (\text{by } t \text{ formula}) \\ 2\theta &= 2k\pi \pm \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) \\ \theta &= k\pi \pm \frac{1}{2} \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) \quad \text{but } -\frac{\pi}{2} < \theta < \frac{\pi}{2} \\ &= \pm \frac{1}{2} \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\end{aligned}$$

When $x \geq 0$ then $0 \leq \theta < \frac{\pi}{2} \Rightarrow 0 \leq 2\theta < \pi$ which aligns with the range of $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, hence for $x \geq 0$

$$\theta = \frac{1}{2} \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right).$$

When $x < 0$ then $-\frac{\pi}{2} < \theta < 0 \Rightarrow -\pi < 2\theta < 0$ which aligns with the range of $-\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, hence for $x < 0$

$$\theta = -\frac{1}{2} \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right).$$

Altogether, this means

$$\tan^{-1} x = \begin{cases} -\frac{1}{2} \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) & \text{for } x < 0 \\ \frac{1}{2} \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) & \text{for } x \geq 0. \end{cases}$$

- (b) (i) The parametric equations of the circle $x^2 + y^2 = r^2$ for some parameter θ are

$$\begin{aligned}x &= r \cos \theta \\y &= r \sin \theta \\\frac{dx}{d\theta} &= -r \sin \theta \\&= -y \\\frac{dy}{d\theta} &= r \cos \theta \\&= x\end{aligned}$$

Differentiate both sides of the given property with respect to θ , noting that r is a constant and independent of θ .

$$\begin{aligned}g(r) &= f(x)f(y) \\\frac{d}{d\theta} [f(x)f(y)] &= 0 \\f(x)\frac{d}{d\theta}f(y) + f(y)\frac{d}{d\theta}f(x) &= 0 \\f(x)\frac{d}{dy}f(y)\frac{dy}{d\theta} + f(y)\frac{d}{dx}f(x)\frac{dx}{d\theta} &= 0 \\f(x)f'(y)x - f(y)f'(x)y &= 0 \\\frac{f'(x)}{xf(x)} &= \frac{f'(y)}{yf(y)}\end{aligned}$$

Noting that $f(x) > 0$ and $f(y) > 0$ as $f(z)$ is a probability density function. Also, note that $x \neq 0$ and $y \neq 0$.

- (ii) The LHS and RHS of the result in part (i) are two identical expressions except x is interchanged with y . Since this is true for all (x, y) on the circle $x^2 + y^2 = r^2$ then it must be that they must equal a constant value.

However, note that $f(x)$ has a global maximum turning point at $x = 0$. This means that when $x < 0$ then $f'(x) > 0$ which implies $xf'(x) < 0$. Similarly when $x > 0$ then $f'(x) < 0$ which also implies $xf'(x) < 0$. Since $f(x) > 0$ for all x (because it is a probability density function) then the constant must be negative since it is a positive value divided by negative value.

Hence, $\frac{f'(x)}{xf(x)} = -k$ for some constant $k > 0$.

(iii) Solving the differential equation

$$\begin{aligned}\frac{f'(x)}{xf(x)} &= -k \\ \frac{f'(x)}{f(x)} &= -kx \\ \int \frac{f'(x)}{f(x)} dx &= -k \int x dx \\ \ln f(x) &= -\frac{kx^2}{2} + c \quad \text{when } x = 0, f(x) = 1 \Rightarrow c = 0 \\ f(x) &= e^{-\frac{kx^2}{2}}\end{aligned}$$

The final step is to find a specific value for the constant k . Since $f(x)$ is a probability density function on all real x then

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) dx &= 1 \\ \int_{-\infty}^{\infty} e^{-\frac{kx^2}{2}} dx &= 1\end{aligned}$$

Let $u = x\sqrt{k} \Rightarrow du = \sqrt{k} dx$. When $x \rightarrow \infty$ then $u = \infty$ and when $x \rightarrow -\infty$ then $u \rightarrow -\infty$ so

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2}} dx &= \frac{1}{\sqrt{k}} \int_{-\infty}^{\infty} e^{-\frac{u^2}{2}} du \\ 1 &= \frac{1}{\sqrt{k}} \int_{-\infty}^{\infty} e^{-\frac{u^2}{2}} du\end{aligned}$$

However, recall that for a standard normal probability density function

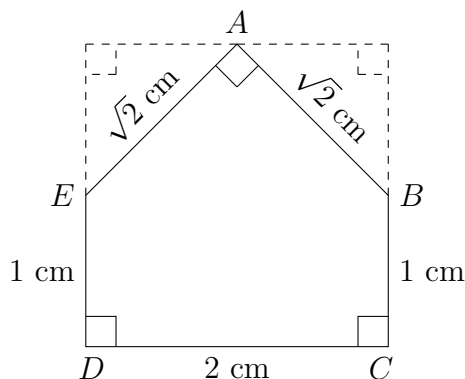
$$\int_{-\infty}^{\infty} \frac{e^{-\frac{u^2}{2}}}{\sqrt{2\pi}} du = 1$$

Hence, $k = 2\pi$ so the specific solution is $f(x) = e^{-\pi x^2}$.

- (c) (i) Fix any random point inside the pentagon as the first point. If the second point does not lie on perimeter of the pentagon, then it is always possible to increase the distance to the fixed point by moving that second point further out towards the pentagon's perimeter. Noting that the first and second points are interchangeable, then this means the longest distance between any two points must be between two points on the perimeter of the pentagon.

Now consider if the first point is fixed on an edge of the pentagon. If the second point also lies on an edge then it is always possible to increase the distance further by moving along the edge in a certain direction. This does not necessarily apply if that second point lies on a vertex instead. Noting that the first and second points are interchangeable, then this means the longest distance must be between any two vertices. In particular, non-adjacent vertices because the distance can always be increased if the points lie on two adjacent vertices.

- (ii) Label the vertices A, B, C, D and E as below.



Consider all the possible pairwise distances between non-adjacent vertices.

$$AC = \sqrt{2^2 + 1^2} = \sqrt{5} \text{ cm}$$

$$AD = \sqrt{2^2 + 1^2} = \sqrt{5} \text{ cm}$$

$$BD = \sqrt{1^2 + 2^2} = \sqrt{5} \text{ cm}$$

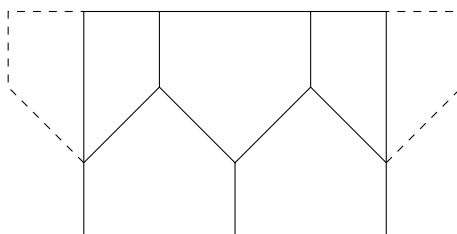
$$BE = 2 \text{ cm}$$

$$CA = \sqrt{2^2 + 1^2} = \sqrt{5} \text{ cm}$$

$$CE = \sqrt{1^2 + 2^2} = \sqrt{5} \text{ cm}$$

Note that by symmetry there are similar results when considering DB, DA, EB and EC . From the above, the longest distance between any two points within the pentagon is $\sqrt{5}$ cm.

- (iii) Split the 3 cm by 4 cm rectangle into the five regions below using the pentagon from part (ii).



When six points are placed at random, at least two must exist in the same region by the pigeonhole principle. In part (ii), it was shown that the largest possible distance between any two points of such a region was $\sqrt{5}$ cm. This means that the distance between these two points in the same region are within $\sqrt{5}$ cm of each other.

(d) When $n = 1$

$$\begin{aligned}
\text{LHS} &= \frac{\sin x}{\cos x + \cos 3x} \\
&= \frac{\sin(2x - x)}{\cos(2x - x) + \cos(2x + x)} \\
&= \frac{\sin 2x \cos x - \cos 2x \sin x}{2 \cos x \cos 2x} \\
&= \frac{\sin 2x}{2 \cos 2x} - \frac{\sin x}{2 \cos x} \\
&= \frac{\tan 2x - \tan x}{2} \\
&= \text{RHS}
\end{aligned}$$

Hence, the statement is true for $n = 1$.

Assume the statement is true for $n = k$.

$$\frac{\sin x}{\cos x + \cos 3x} + \frac{\sin x}{\cos x + \cos 5x} + \cdots + \frac{\sin x}{\cos x + \cos(2k + 1)x} = \frac{\tan(k + 1)x - \tan x}{2}$$

Required to prove the statement is true for $n = k + 1$.

$$\frac{\sin x}{\cos x + \cos 3x} + \frac{\sin x}{\cos x + \cos 5x} + \cdots + \frac{\sin x}{\cos x + \cos(2k + 3)x} = \frac{\tan(k + 2)x - \tan x}{2}$$

$$\begin{aligned}
\text{LHS} &= \frac{\sin x}{\cos x + \cos 3x} + \frac{\sin x}{\cos x + \cos 5x} + \cdots + \frac{\sin x}{\cos x + \cos(2k + 1)x} + \frac{\sin x}{\cos x + \cos(2k + 3)x} \\
&= \frac{\tan(k + 1)x - \tan x}{2} + \frac{\sin x}{\cos x + \cos(2k + 3)x} \quad (\text{by assumption}) \\
&= \frac{\tan(k + 1)x}{2} + \frac{\sin x}{\cos[(k + 2) - (k + 1)]x + \cos[(k + 2) + (k + 1)]x} - \frac{\tan x}{2} \\
&= \frac{\sin(k + 1)x}{2 \cos(k + 1)x} + \frac{\sin x}{2 \cos(k + 2)x \cos(k + 1)x} - \frac{\tan x}{2} \\
&= \frac{\sin(k + 1)x \cos(k + 2)x + \sin x}{2 \cos(k + 2)x \cos(k + 1)x} - \frac{\tan x}{2} \\
&= \frac{\sin(k + 1)x \cos(k + 2)x + \sin[(k + 2) - (k + 1)]x}{2 \cos(k + 2)x \cos(k + 1)x} - \frac{\tan x}{2} \\
&= \frac{\sin(k + 1)x \cos(k + 2)x + \sin(k + 2)x \cos(k + 1)x - \cos(k + 2)x \sin(k + 1)x}{2 \cos(k + 2)x \cos(k + 1)x} - \frac{\tan x}{2} \\
&= \frac{\sin(k + 2)x \cos(k + 1)x}{2 \cos(k + 2)x \cos(k + 1)x} - \frac{\tan x}{2} \\
&= \frac{\tan(k + 2)x - \tan x}{2} \\
&= \text{RHS}
\end{aligned}$$

Since the statment is true for $n = 1$, then by induction it is true all positive integers n .

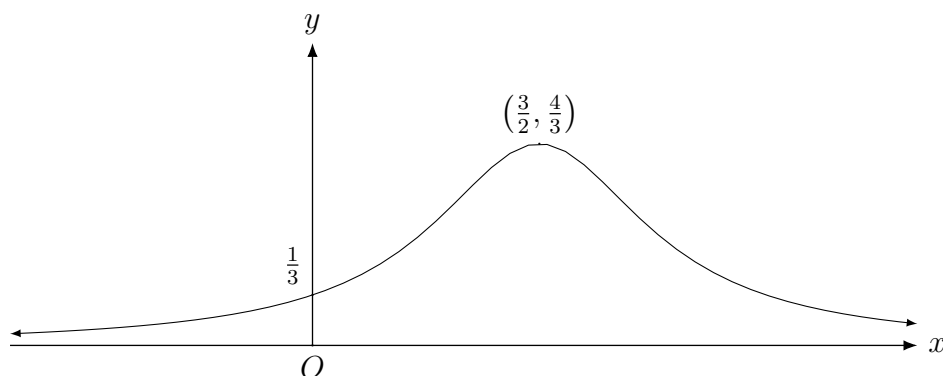
Question 14

- (a) (i) First consider $y = x^2 - 3x + 3$. It's discriminant is $\Delta = -3 < 0$ so it is positive definite. It has a minimum turning point at $(\frac{3}{2}, \frac{3}{4})$.

For the graph of $y = \frac{1}{x^2 - 3x + 3}$, when $x \rightarrow \pm\infty$ then $y \rightarrow 0$.

Since $x^2 - 3x + 3$ is positive definite then $y > 0$.

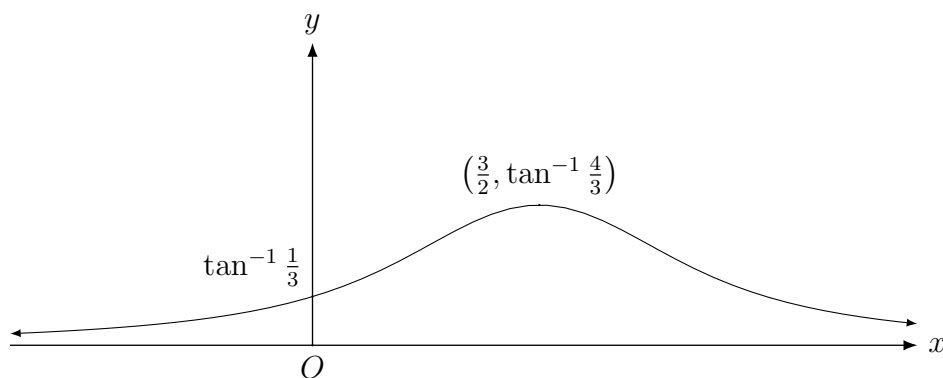
It also has a maximum turning point at $(\frac{3}{2}, \frac{4}{3})$.



- (ii) For the graph of $y = \tan^{-1}\left(\frac{1}{x^2 - 3x + 3}\right)$, when $x \rightarrow \pm\infty$ then $\frac{1}{x^2 - 3x + 3} \rightarrow 0$ so $y \rightarrow 0$.

Since $x^2 - 3x + 3$ is positive definite then $\frac{1}{x^2 - 3x + 3} > 0$ so $y > 0$.

It also has a maximum turning point at $(\frac{3}{2}, \tan^{-1} \frac{4}{3})$.



- (iii) Consider the graphs of $f(x) = x$ and $g(x) = \tan^{-1} x$.

$$\begin{aligned} f'(x) &= 1 \\ g'(x) &= \frac{1}{1+x^2} \end{aligned}$$

For $x > 0$, $f'(x) > g'(x)$. Since $f'(0) = g'(0) = 1$ then $f(x) > g(x)$ for $x > 0$. Hence, $x > \tan^{-1} x$ for $x > 0$.

Note that since $\frac{1}{x^2 - 3x + 3} > 0$ then the equivalent argument applies that $\frac{1}{x^2 - 3x + 3} > \tan^{-1} \left(\frac{1}{x^2 - 3x + 3} \right)$. Hence, the graphs never intersect.

- (iv) Differentiating with respect to x

$$\begin{aligned} & \frac{d}{dx} \left[(x+c) \tan^{-1}(x+c) - \ln \left(\sqrt{1+(x+c)^2} \right) \right] \\ &= \frac{d}{dx} \left[(x+c) \tan^{-1}(x+c) - \frac{1}{2} \ln(1+(x+c)^2) \right] \\ &= \tan^{-1}(x+c) + \frac{x+c}{1+(x+c)^2} - \frac{1}{2} \times \frac{2(x+c)}{1+(x+c)^2} \\ &= \tan^{-1}(x+c) \end{aligned}$$

- (v) First note that

$$\begin{aligned} \frac{1}{x^2 - 3x + 3} &= \frac{1}{x^2 - 3x + 2 + 1} \\ &= \frac{1}{(x-1)(x-2) + 1} \\ &= \frac{x-1+2-x}{1-(x-1)(2-x)} \end{aligned}$$

Let $\alpha = \tan^{-1}(x-1)$ and $\beta = \tan^{-1}(2-x)$ so

$$\begin{aligned} \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ &= \frac{x-1+2-x}{1-(x-1)(2-x)} \\ &= \frac{1}{x^2 - 3x + 3} \\ \alpha + \beta &= \tan^{-1} \left(\frac{1}{x^2 - 3x + 3} \right) \\ \tan^{-1}(x-1) + \tan^{-1}(2-x) &= \tan^{-1} \left(\frac{1}{x^2 - 3x + 3} \right) \end{aligned}$$

Let A be the area between the curves.

$$\begin{aligned}
A &= \int_1^2 \frac{dx}{x^2 - 3x + 3} - \int_1^2 \tan^{-1} \left(\frac{1}{x^2 - 3x + 3} \right) dx \\
&= \int_1^2 \frac{dx}{x^2 - 3x + \frac{9}{4} + \frac{3}{4}} - \int_1^2 \tan^{-1}(x - 1) dx - \int_1^2 \tan^{-1}(2 - x) dx \\
&= \int_1^2 \frac{dx}{\left(x - \frac{3}{2}\right)^2 + \frac{3}{4}} - \int_1^2 \tan^{-1}(x - 1) dx + \int_1^2 \tan^{-1}(x - 2) dx
\end{aligned}$$

Evaluating each integral individually and using the result in part (iv)

$$\begin{aligned}
\int_1^2 \frac{dx}{\left(x - \frac{3}{2}\right)^2 + \frac{3}{4}} &= \frac{2}{\sqrt{3}} \left[\tan^{-1} \left(\frac{2\left(x - \frac{3}{2}\right)}{\sqrt{3}} \right) \right]_1^2 \\
&= \frac{2\pi}{3\sqrt{3}} \\
\int_1^2 \tan^{-1}(x - 1) dx &= \left[(x - 1) \tan^{-1}(x - 1) - \ln \left(\sqrt{1 + (x - 1)^2} \right) \right]_1^2 \\
&= \frac{\pi}{4} - \ln \sqrt{2} \\
\int_1^2 \tan^{-1}(x - 2) dx &= \left[(x - 2) \tan^{-1}(x - 2) - \ln \left(\sqrt{1 + (x - 2)^2} \right) \right]_1^2 \\
&= -\frac{\pi}{4} + \ln \sqrt{2} \\
A &= \frac{2\pi}{3\sqrt{3}} - \frac{\pi}{2} + 2 \ln \sqrt{2}
\end{aligned}$$

- (b) To r th goal is fixed as the last goal which ends the training session. The soccer player must score $(r - 1)$ goals in the prior games, with a maximum of $(n - 1)$ games allowed.

If she scores $(r - 1)$ goals within $(r - 1)$ games there are $\binom{r - 1}{r - 1}$ possible ways.

If she scores $(r - 1)$ goals within r games there are $\binom{r}{r - 1}$ possible ways.

...

If she scores $(r - 1)$ goals within $(n - 1)$ games there are $\binom{n - 1}{r - 1}$ possible ways.

The total number of possibilities is the LHS.

However, this can also be interpreted as arranging r goals within n games in any order. The last goal must always be the end of the training session and the subsequent 'non-goals' can be ignored, as this does not change the total number of possibilities. There are $\binom{n}{r}$ possibilities, which is the RHS. Hence,

$$\binom{r - 1}{r - 1} + \binom{r}{r - 1} + \binom{r + 1}{r - 1} + \cdots + \binom{n - 1}{r - 1} = \binom{n}{r}.$$

(c) (i) For integer $n \geq 4$

$$\begin{aligned}
\text{RHS} &= 24\binom{n}{4} + 36\binom{n}{3} + 14\binom{n}{2} + \binom{n}{1} \\
&= \frac{24n!}{4!(n-4)!} + \frac{36n!}{3!(n-3)!} + \frac{14n!}{2!(n-2)!} + \frac{n!}{1!(n-1)!} \\
&= n(n-1)(n-2)(n-3) + 6n(n-1)(n-2) + 7n(n-1) + n \\
&= (n^2 - n)(n^2 - 5n + 6) + 6n(n^2 - 3n + 2) + 7n^2 - 7n + n \\
&= n^4 - 5n^3 + 6n^2 - n^3 + 5n^2 - 6n + 6n^3 - 18n^2 + 12n + 7n^2 - 7n + n \\
&= n^4 \\
&= \text{LHS}
\end{aligned}$$

(ii) Using the result in part (c)(i)

$$\begin{aligned}
4^4 &= 24\binom{4}{4} + 36\binom{4}{3} + 14\binom{4}{2} + \binom{4}{1} \\
5^4 &= 24\binom{5}{4} + 36\binom{5}{3} + 14\binom{5}{2} + \binom{5}{1} \\
6^4 &= 24\binom{6}{4} + 36\binom{6}{3} + 14\binom{6}{2} + \binom{6}{1} \\
&\dots \\
n^4 &= 24\binom{n}{4} + 36\binom{n}{3} + 14\binom{n}{2} + \binom{n}{1}
\end{aligned}$$

Also, from part (b)

$$\begin{aligned}
\binom{n+1}{5} &= \binom{4}{4} + \binom{5}{4} + \binom{6}{4} + \dots + \binom{n}{4} \\
\binom{n+1}{4} &= \binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \dots + \binom{n}{3} \\
\binom{n+1}{3} &= \binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \dots + \binom{n}{2} \\
\binom{n+1}{2} &= \binom{1}{1} + \binom{2}{1} + \binom{3}{1} + \dots + \binom{n}{1}
\end{aligned}$$

Applying these results

LHS

$$\begin{aligned}
&= 1^4 + 2^4 + 3^4 + 4^4 + 5^4 + 6^4 + \dots + n^4 \\
&= 1^4 + 2^4 + 3^4 + 24 \binom{n+1}{5} + 36 \left[\binom{n+1}{4} - \binom{3}{3} \right] \\
&\quad + 14 \left[\binom{n+1}{3} - \binom{3}{2} - \binom{2}{2} \right] + \binom{n+1}{2} - \binom{3}{1} - \binom{2}{1} - \binom{1}{1} \\
&= 24 \binom{n+1}{5} + 36 \binom{n+1}{4} + 14 \binom{n+1}{3} + \binom{n+1}{2} \\
&\quad + 1^4 + 2^4 + 3^4 - 36 \binom{3}{3} - 14 \binom{3}{2} - 14 \binom{2}{2} - \binom{3}{1} - \binom{2}{1} - \binom{1}{1} \\
&= 24 \binom{n+1}{5} + 36 \binom{n+1}{4} + 14 \binom{n+1}{3} + \binom{n+1}{2} \\
&= \frac{24(n+1)!}{5!(n-4)!} + \frac{36(n+1)!}{4!(n-3)!} + \frac{14(n+1)!}{3!(n-2)!} + \frac{(n+1)!}{2!(n-1)!} \\
&= \frac{(n+1)n(n-1)(n-2)(n-3)}{5} + \frac{3(n+1)n(n-1)(n-2)}{2} \\
&\quad + \frac{7(n+1)n(n-1)}{3} + \frac{(n+1)n}{2} \\
&= \frac{1}{30} n(n+1) [6(n-1)(n-2)(n-3) + 45(n-1)(n-2) + 70(n-1) + 15] \\
&= \frac{1}{30} n(n+1) [6(n-1)(n^2 - 5n + 6) + 45(n^2 - 3n + 2) + 70n - 70 + 15] \\
&= \frac{1}{30} n(n+1) [6n^3 - 30n^2 + 36n - 6n^2 + 30n - 36 + 45n^2 - 135n + 90 + 70n - 70 + 15] \\
&= \frac{1}{30} n(n+1) (6n^3 + 9n^2 + n - 1) \\
&= \frac{1}{30} n(n+1) (6n^3 + 9n^2 + 3n - 2n - 1) \\
&= \frac{1}{30} n(n+1) (3n(2n^2 + 3n + 1) - (2n + 1)) \\
&= \frac{1}{30} n(n+1) (3n(2n+1)(n+1) - (2n+1)) \\
&= \frac{1}{30} n(n+1) (2n+1) (3n(n+1) - 1) \\
&= \frac{1}{30} n(n+1) (2n+1) (3n^2 + 3n - 1) \\
&= \text{RHS}
\end{aligned}$$